GitHub Authentication & Git Remote Access

Part I: Configure two-factor authentication for GitHub

GitHub now **requires** two-factor authentication (2FA) for most active contributors. Before you configure 2FA, you login to your GitHub account using your username and password. After you configure 2FA, it takes one more step to login to your GitHub account.

There are multiple ways to do it. Today we introduce one way.

Step 1. Install a TOTP app

First, you need to install a time-based one-time password (TOTP) application on your phone or on your desktop.

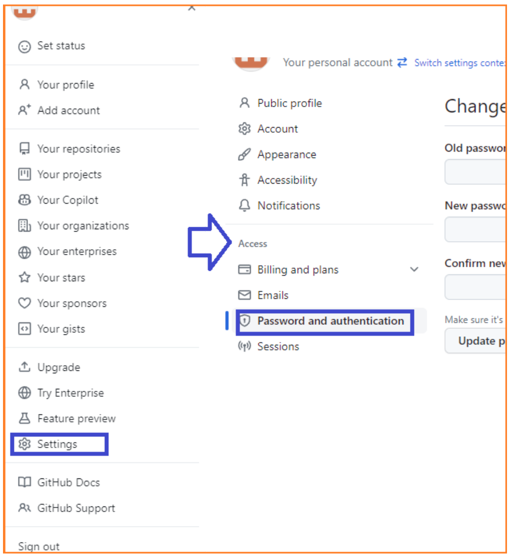
You may choose Microsoft Authenticator to do it. The software can be found on some app stores. However, we have no requirement on what you use.

Step 2. Set up 2FA on GitHub

After installing a TOTP app, we can configure GitHub to use it.

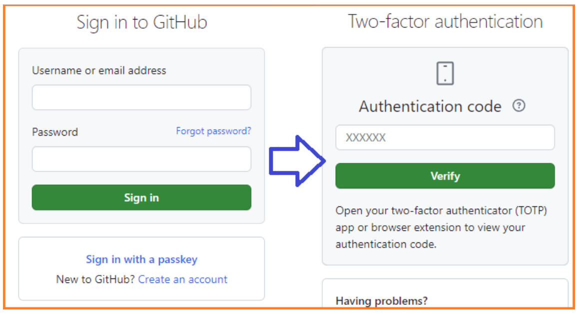
Login to Github using username and password, click "Settings", "Password and authentication", scroll down to "Two-factor authenication", click "Enable two-factor authentication".

Then you will see a QR code. Now open your TOTP app you have just installed to scan the QR code and follow the steps on your app.



Step 3. GitHub Login

In the future, when you need to login to your GitHub account, GitHub will first ask for your username and password as usual. Then GitHub will ask for a 6-digit authetication code. Open your TOTP app, click your GitHub account on the app (it is there because you just scanned that QR code). You will see the 6-digit code. Type this code and you can login.



Step 4. Save Recovery code

Just in case you lost your phone and won't be able to login your GitHub account forever, you may want to download your recovery code.

Part II: Remote Access via HTTPS

HTTPS is beginner-friendly and requires no SSH key setup.

Step 1. Copy HTTPS URL

Navigate to your repository on GitHub, click the green **Code** button, select the **HTTPS** tab, click the copy icon to copy the URL (e.g., https://github.com/`<username>`/Teedy.git)

https

Step 2. Clone the Repository

Open your terminal(Windows: Open Command Prompt or PowerShell; Mac/Linux: Open Terminal.), navigate to your desired directory.

```
cd <Your Desired Folder>
```

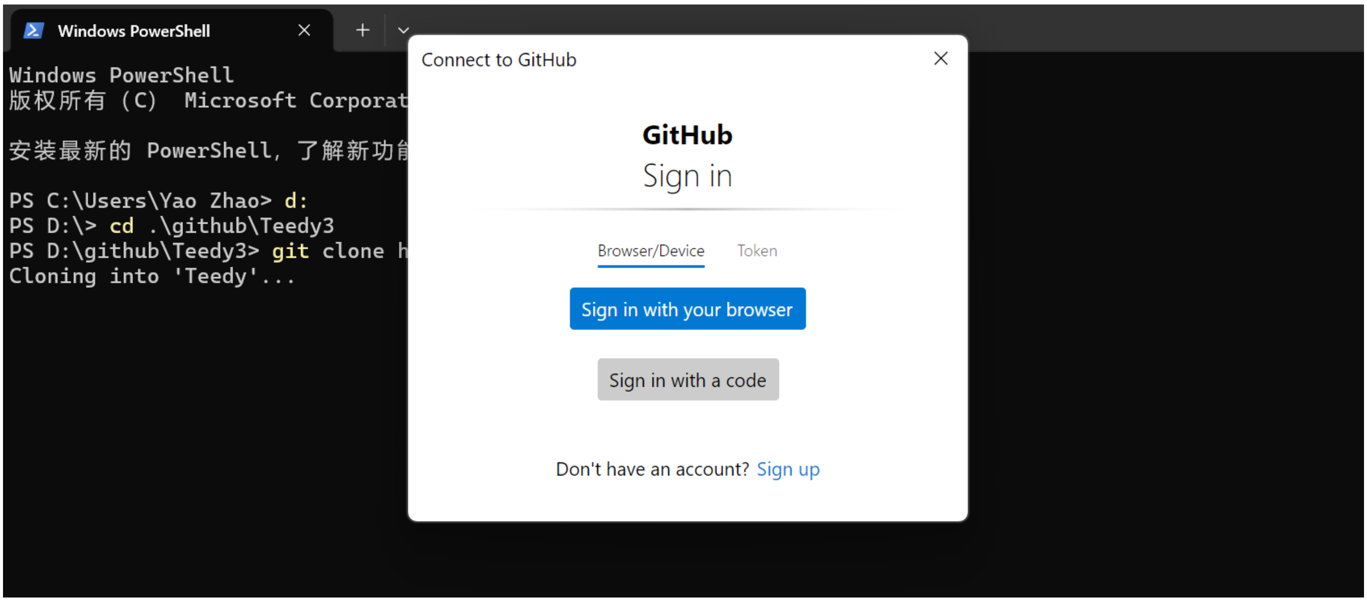
Run the clone command:

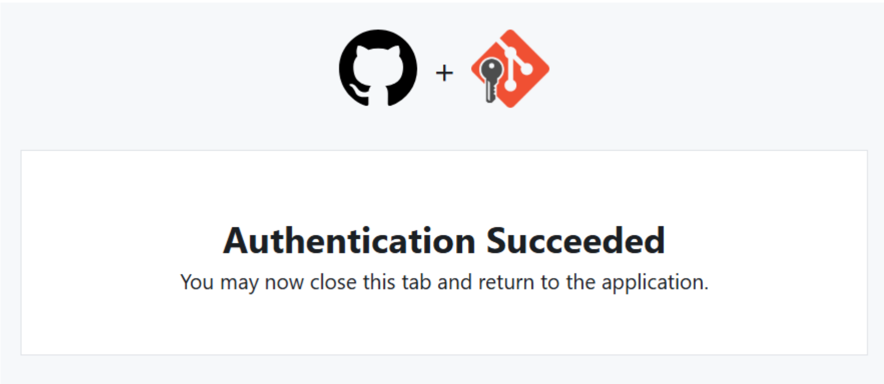
```
git clone https://github.com/<username>/Teedy.git
```

Replace the URL with your actual repository link.

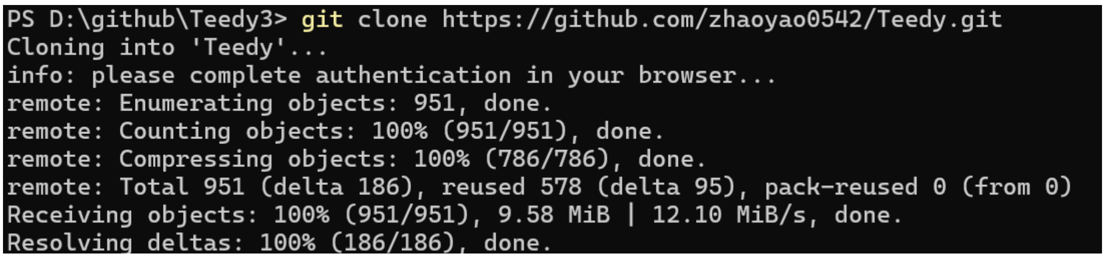
Step 3: Authentication

Git will pop up a window, you can select `Sign in with your browser`. Git will open a browser window, redirect to GitHub login. You can follow the login process to complete the verification.





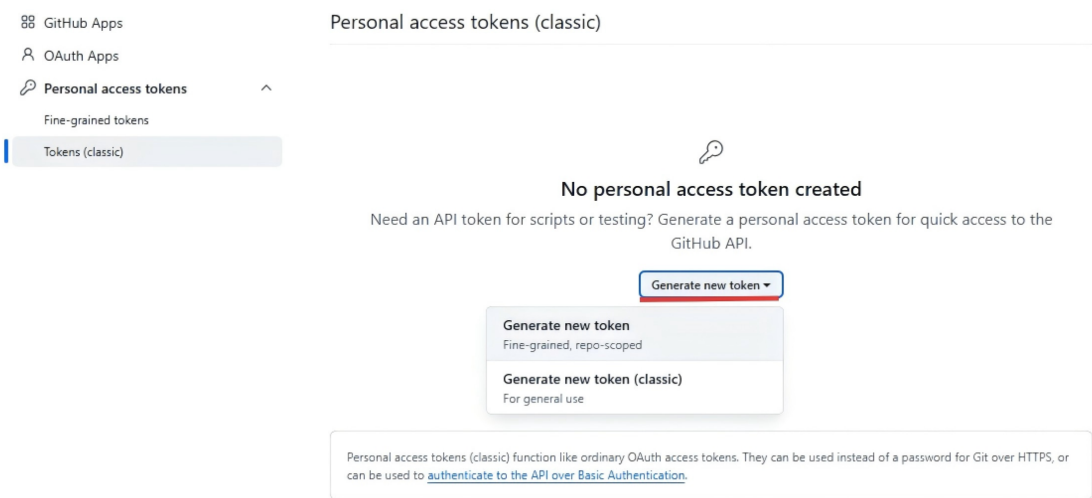
After successful login, Git will receive an OAuth token, operation continues.

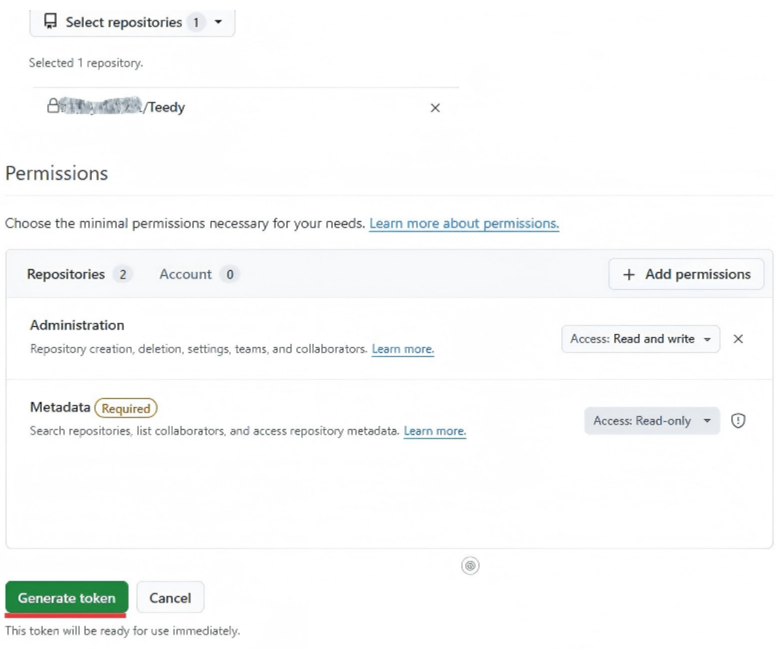


```
PS D:\github\Teedy3> git clone https://github.com/zhaoyao0542/Teedy.git
Cloning into 'Teedy'...
info: please complete authentication in your browser...
remote: Enumerating objects: 951, done.
remote: Counting objects: 100% (951/951), done.
remote: Compressing objects: 100% (786/786), done.
remote: Total 951 (delta 186), reused 578 (delta 95), pack-reused 0 (from 0)
Receiving objects: 100% (951/951), 9.58 MiB | 12.10 MiB/s, done.
Resolving deltas: 100% (186/186), done.
```

If browser auth fails, use a Personal Access Token (PAT):

1. Generate a token with `repo` scope



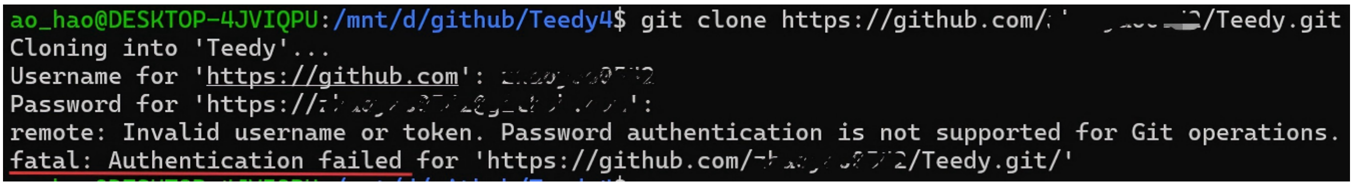


2. When prompted for password, paste the token (not your account password)

```
Username: your_username
Password: github_pat_xxxxxxxxxxxxxxxxxxxx
```

Part III: Remote Access via SSH

On Linux, HTTPS is more likely to encounter problems.



We recommend using SSH instead. SSH does not require repeated token or password entry, is better suited for long-term, frequent Git users.

Step 1: Generate SSH Key

In terminal:

```
ssh-keygen -t rsa -C "your_email@example.com"
```

Press Enter for default path.

This generates:

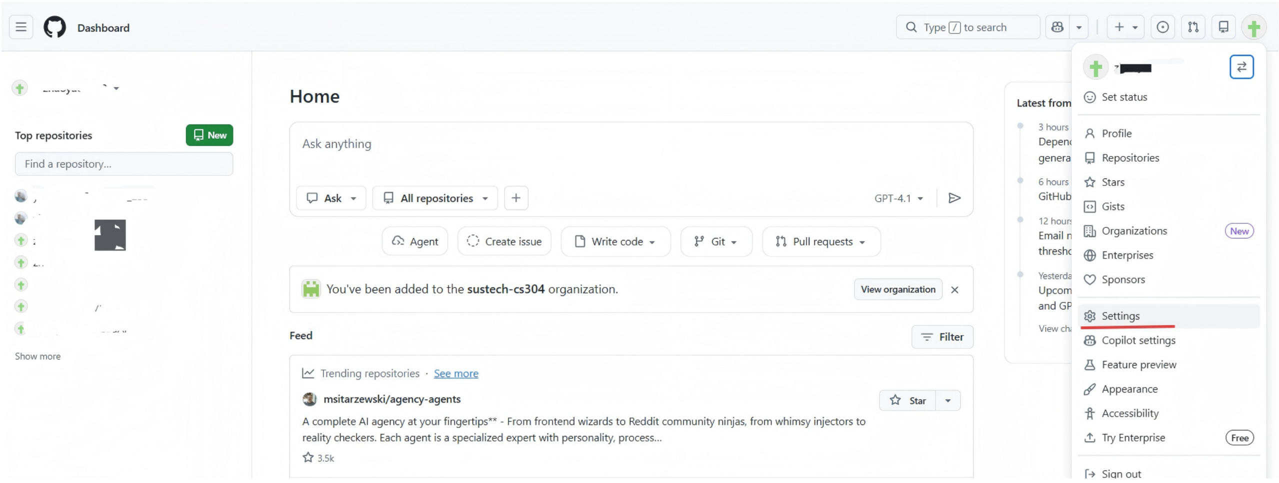
- Private key: `~/.ssh/rsa`
- Public key: `~/.ssh/rsa.pub`

Step 2: Add SSH Key to GitHub

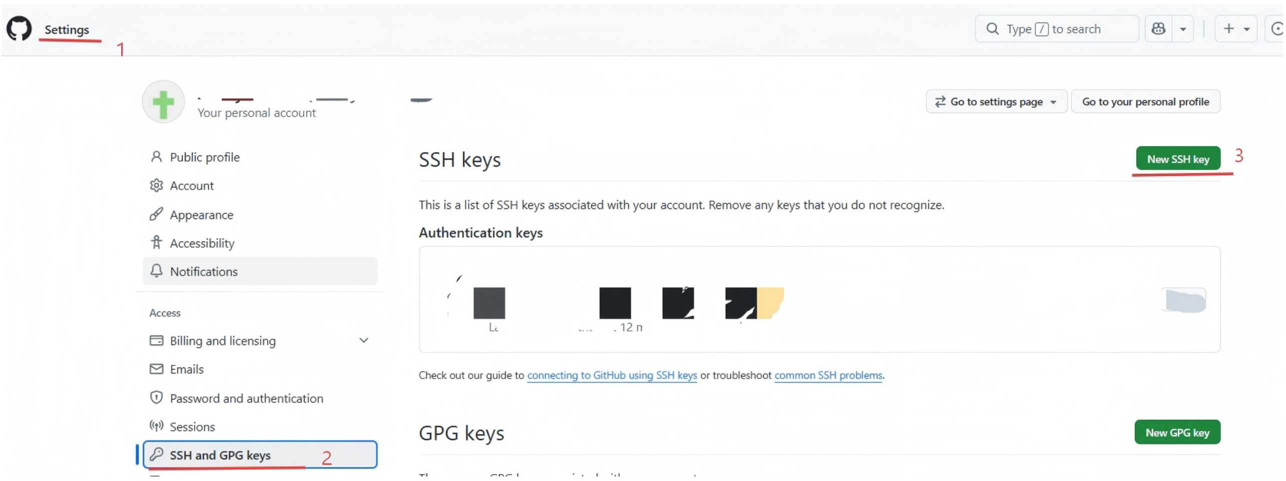
1. Copy public key:

```
cat ~/.ssh/rsa.pub
```

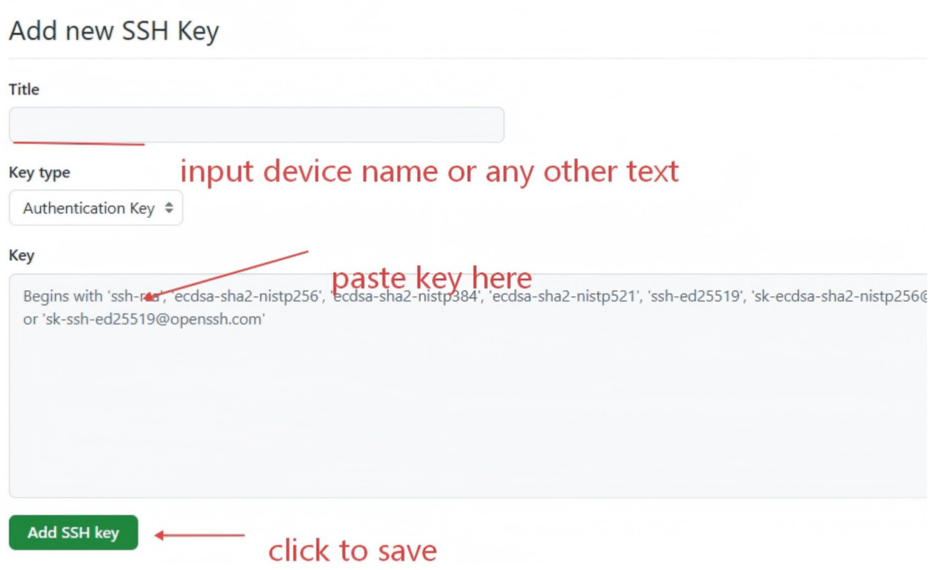
2. Go to GitHub → **Settings** → **SSH and GPG keys**



3. Click **New SSH key**



4. Paste key and save



Step 3: Test SSH Connection

```
ssh -T git@github.com
```

If successful:

```
Hi username! You've successfully authenticated...
```

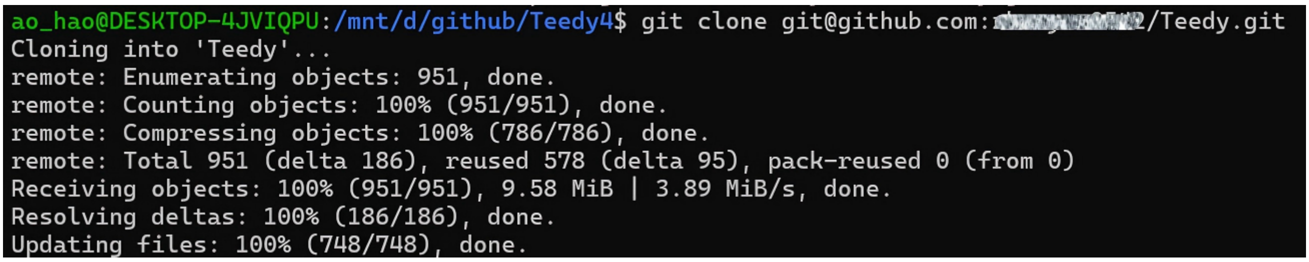
Step 4: Clone Using SSH

Copy SSH URL:

```
git@github.com:username/repository.git
```

Then:

```
git clone git@github.com:username/repository.git
```



```
ao_hao@DESKTOP-4JVIQPU:/mnt/d/github/Teedy4$ git clone git@github.com:[redacted]/Teedy.git
Cloning into 'Teedy'...
remote: Enumerating objects: 951, done.
remote: Counting objects: 100% (951/951), done.
remote: Compressing objects: 100% (786/786), done.
remote: Total 951 (delta 186), reused 578 (delta 95), pack-reused 0 (from 0)
Receiving objects: 100% (951/951), 9.58 MiB | 3.89 MiB/s, done.
Resolving deltas: 100% (186/186), done.
Updating files: 100% (748/748), done.
```

Reference:

[Configuring two-factor authentication]
[Configuring two-factor authentication recovery methods]
[Generating a new SSH key and adding it to the ssh-agent]